

11FIT WhatsApp AI Assistant

Capacity, Throughput & Cost Architecture Report | Groq Llama-3.3-70B Engine

100% FREE TIER
ENABLED

1. Executive Summary & Free Tier Quotas

The 11FIT WhatsApp AI Assistant is powered by **Groq's Llama-3.3-70B-Versatile** model running on specialized Language Processing Units (LPUs). This provides ultra-low latency (~288ms response times) along with a generous daily developer free quota that requires no upfront payment or credit card.

DAILY MESSAGE CAPACITY (FREE) 800 - 1,000 Customer messages answered per day	DAILY CUSTOMER CHAT SESSIONS 200 - 250 Unique customers chatting daily (Avg 4 msgs)	MONTHLY FREE MESSAGES 25,000+ ₹0.00 infrastructure AI cost
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2. Exact Capacity Breakdown (Daily vs. Monthly)

Metric / Workload Type	Per Message Avg	Daily Limit (Free Tier)	Monthly Capacity (Free)	Status
Token Utilization	~600 Tokens (In + Out)	500,000 Tokens / day	~15,000,000 Tokens / mo	100% FREE
API Request Limit	1 Request / reply	14,400 Requests / day	~432,000 Requests / mo	100% FREE
Customer Message Replies	1 AI Reply	800 - 1,000 Replies/day	25,000 - 30,000 / mo	ACTIVE CAPACITY
Unique Customer Chats	4 Messages / session	200 - 250 Customers/day	6,000 - 7,500 / mo	ACTIVE CAPACITY
Peak Speed / Throughput	~288 Milliseconds	30 Requests / minute	Instant Auto-scaling	HIGH SPEED

3. Scale-Up & Exceeding Free Quota (Cost Modeling)

If your brand grows beyond **30,000 WhatsApp replies per month**, you can either switch API keys instantly via the app UI or upgrade to Groq Pay-As-You-Go. Groq's token pricing is among the lowest in the industry:

Monthly Growth Tier	Total Monthly Messages	Total Tokens Used	Estimated Monthly Cost (USD)	Estimated Monthly Cost (INR)
Tier 1 (Current Free)	Up to 25,000 messages	15 Million Tokens	\$0.00	₹0 INR (FREE)
Tier 2 (High Growth)	50,000 messages	30 Million Tokens	~\$10.35 USD	~₹860 INR / mo
Tier 3 (Enterprise)	100,000 messages	60 Million Tokens	~\$20.70 USD	~₹1,720 INR / mo

4. 3-Model High-Availability Fallback Chain

To guarantee zero downtime even during traffic spikes, the webhook enforces an automated fallback sequence:

Priority Order	Model Identifier	Avg Response Time	Role & Behavior
1. Primary Engine	llama-3.3-70b-versatile	~288 ms	Highest intelligence, nuanced Hinglish & Hindi styling
2. Instant Fallback	llama-3.1-8b-instant	~123 ms	Ultra-fast backup if 70B is busy or rate-limited
3. Compound Backup	compound-beta	~904 ms	Deep reasoning backup ensuring 99.99% reliability

 **Real-Time API Key Switching via Dashboard**

You never need to redeploy code if you want to change AI keys. Simply open <https://shopify-price-editor.vercel.app> → click **Settings** →  **WhatsApp AI Bot** and paste a new free key. The live webhook updates instantly!